

Pure strategy game theory

- Your outcomes in the world don't just depend on your own actions
- Other people around you are also making decisions/actions that affect you
- Our goal: Find the pairs of actions where no one wants to change their mind
 - We think about this as every player being at their **local and individual** optimal point
 - Compare this to the monopoly maximizing profit, where the monopoly is at the **global** optimal point
- We call these points Nash Equilibria

Pure strategy game theory

It's useful to think about the questions we are asking, and those we aren't asking



We are asking: What are the pairs of decisions so that no individual would want to deviate? That is, what are the Nash Equilibria?



We are not asking: How do I know that the other player is playing its Nash Equilibrium strategy?



We are not asking: If there are multiple Nash Equilibria, which one should I choose?



We are not asking: How do I even know what the other player's payoffs are?

These are all interesting questions – but in this course we're really just focused on finding the Nash Equilibria

Pure strategy game theory

How do we find Nash Equilibria

1. Start with our payoff matrix
2. Set Player 2's decision to play option 1: What is the best decision Player 1 could make?
3. Set Player 2's decision to play option 2: What is the best decision Player 1 could make?
4. Do the same, switching Player 1 and Player 2

Where the best decisions point to the same cell, this is a Nash Equilibrium – it means that both players' best decisions should end up in the same place. Neither player would want to make a different decision

The arrows we draw encode the **Best Response** function of each player

Our firm entry game from last lecture

	Firm 1 enters	Firm 1 does not enter
Firm 2 enters	F1 profit = -1 F2 profit = -1	F1 profit = 0 F2 profit = 1
Firm 2 does not enter	F1 profit = 1 F2 profit = 0	F1 profit = 0 F2 profit = 0

Mixed strategy game theory

- Often the optimal play for you to make is randomizing your moves
- This is true in many real-life games:
 - In poker, there is a game-theory optimal portion of the time that you should **bluff** rather than **fold**. If you don't randomize correctly, other players can get an edge over you
 - In soccer, there is a game-theory optimal portion of the time you should **kick a penalty to the right** rather than the **left**. If you don't randomize correctly, the goalie can save it more often than not
- In these settings, players often know they are playing these games and they **do actually** play their equilibrium mixed strategies
- Here, our Nash Equilibria are when **both players want to randomize**

Mixed strategy game theory

Again, it's useful to think about the questions are asking, and those we aren't asking



We are asking: What are the probabilities both players should make decisions so that both players want to keep randomizing?



We are not asking: How do I know the other player is not just playing a pure strategy game – that is, not randomizing?



We are not asking: How do I know the other player is randomizing correctly?



We are not asking: How do I even know the other player's probabilities they are using?

These are all interesting questions – but in this course we're really just focused on finding the mixed Nash Equilibrium probabilities

Mixed strategy game theory

Let's have a thought experiment of why these mixed strategies arise

Imagine a casino is designing a new roulette game where **red or black is chosen randomly**

The gambler is going to play this game by also choosing **red or black randomly**

If the gambler chooses correctly, they win. The winner gets \$1, the loser loses \$1

Two key questions:

1. With what probability should the casino game show red or black?
2. With what probability should the gambler choose red or black?

Let's reason through this, before solving it properly

Mixed strategy game theory

1 - With what probability should the casino game show red or black?

- If the casino game shows red 75% of the time:
 - The gambler should always choose red because they have a 75% chance of winning
 - The gambler gets an edge here and would not randomize (not a NE!)
- If the casino game shows red 25% of the time:
 - The gambler should always choose black because they have a 75% chance of winning
 - The gambler gets an edge here and would not randomize (not a NE!)
- If the casino game shows red 50% of the time:
 - The gambler should be indifferent between choosing black or red, there's no edge they can gain
 - The casino game having red pop up 50% is part of the mixed strategy Nash Equilibrium

Mixed strategy game theory

2 - With what probability should the gambler choose red or black?

- If the gambler chooses red 75% of the time
 - The casino should design their game to always show black because they have a 75% chance of winning
 - The casino gets an edge here and would not randomize (not a NE!)
- If the gambler chooses red 25% of the time
 - The casino should design their game to always show red because they have a 75% chance of winning
 - The casino gets an edge here and would not randomize (not a NE!)
- If the gambler chooses red 50% of the time
 - The casino should be indifferent between their game showing black or red, there's no edge they can gain
 - The gambler choosing red 50% of the time is part of the mixed strategy Nash Equilibrium

Mixed strategy game theory

So for our mixed strategy Nash Equilibrium, we have two conditions that need to happen at once!

- 1 - The casino game having red pop up 50% is part of the mixed strategy Nash Equilibrium
- 2 - The gambler choosing red 50% of the time is part of the mixed strategy Nash Equilibrium

If either the gambler or casino plays even slightly different probabilities, the other player gets an edge and would not longer randomize

I like to think of our mixed strategies as the probabilities that hold everything in this nice, delicate balance. Even small deviations completely unravel the randomization

Let's solve our casino game properly

	Casino: Red	Casino: Black
	Probability that roulette game chooses red = p_C	Probability that roulette game chooses red = $(1 - p_C)$
Gambler: Red Probability that gambler chooses red = p_G	$C = -1$ $G = 1$	$C = 1$ $B = -1$
Gambler: Black Probability that gambler chooses black = $(1 - p_G)$	$C = 1$ $G = -1$	$C = -1$ $B = 1$

Sequential games and first mover advantage

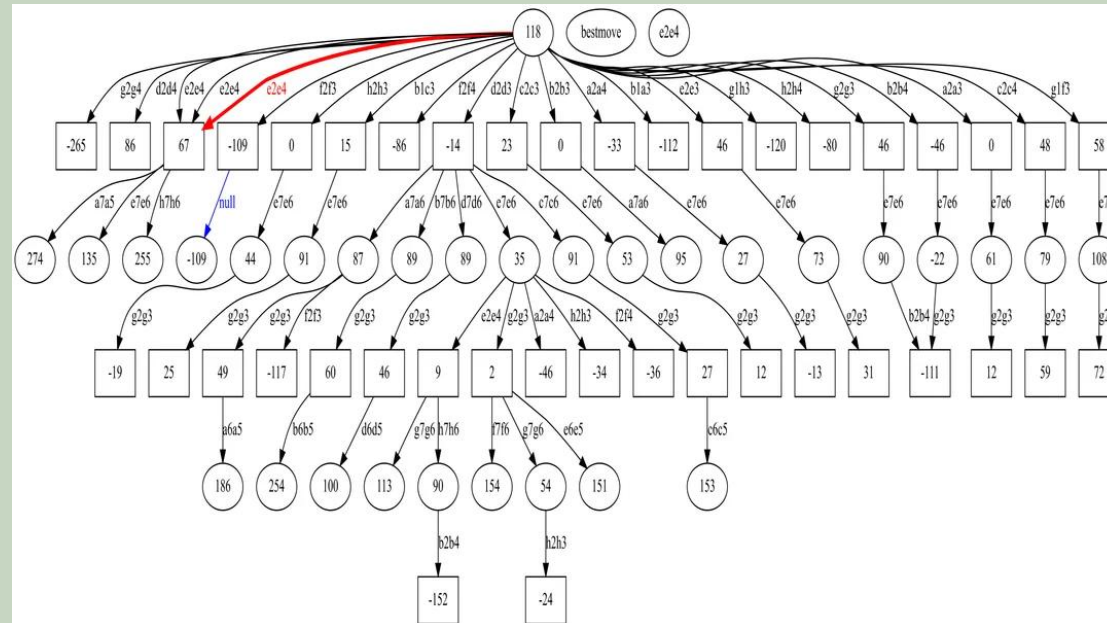
Our games up to now (simultaneous games) rely on each player responding to each other

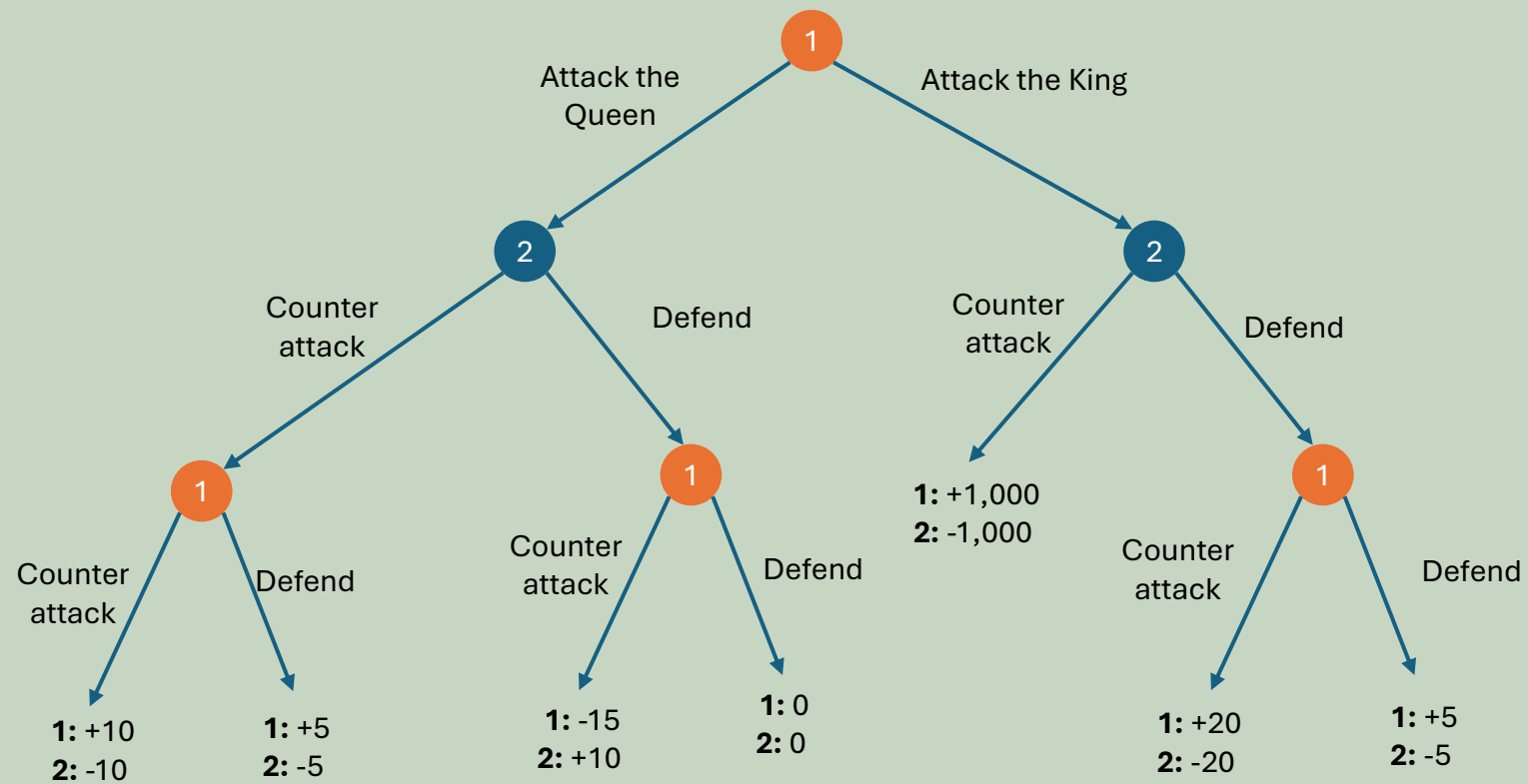
But when Player 1 goes first, they **typically get an advantage** – only Player 2 has to respond to Player 1

Player 1 gets the advantage of anticipating Player 2's best moves

We solve all of the small, subsets of decisions bottom to top to find Player 1's optimal opening moves!

This is literally what chess engines (and chess players!) do when playing games





Lastly, our models of competition

The purpose of our models of competition is to use our tools of game theory to think about how firms compete

Three models:

1. Cournot: Firms compete on choosing quantity/output
2. Bertrand: Firms compete on choosing prices
3. Stackelberg: Firm 1 chooses output first, then Firm 2 responds

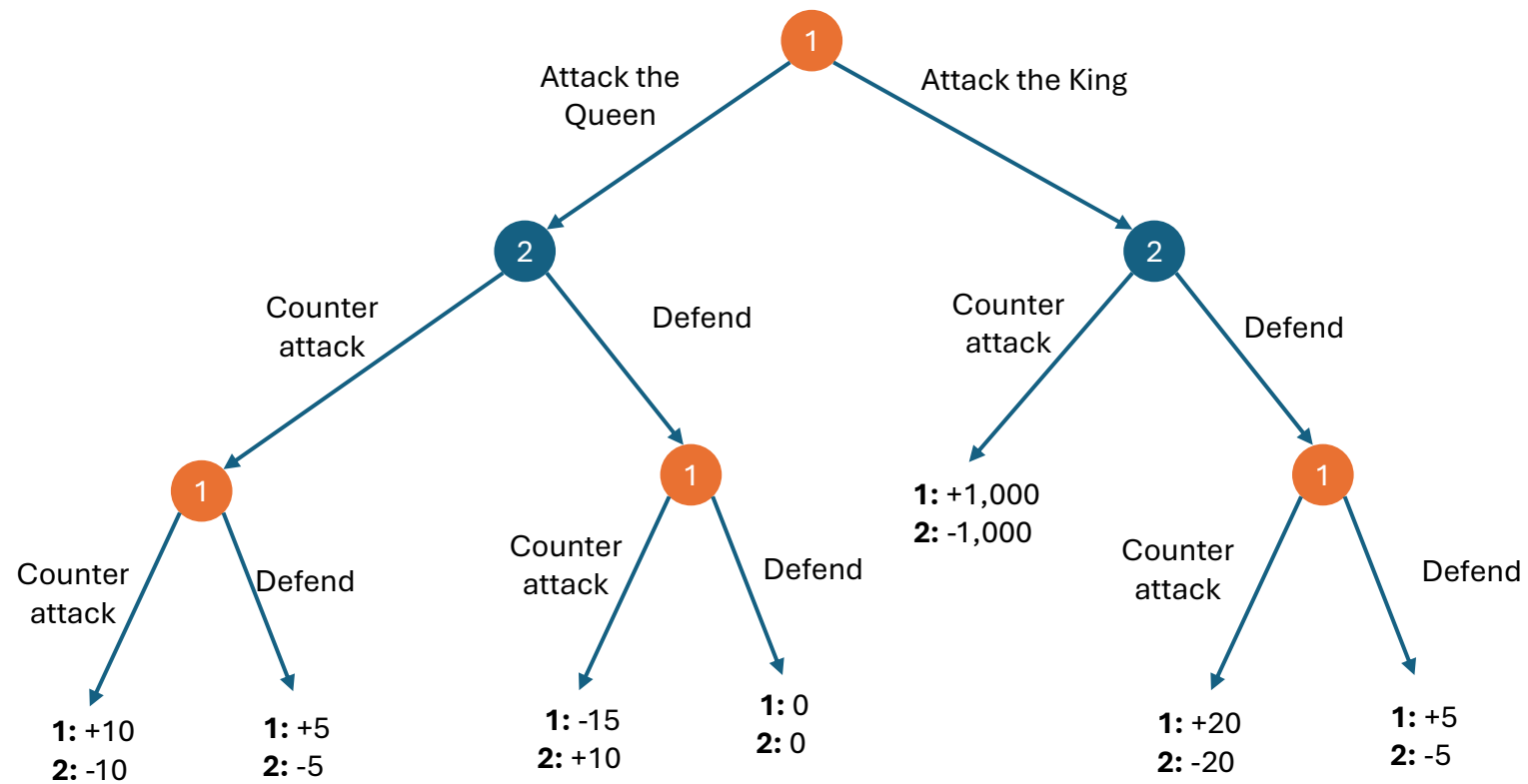
All variations on the same theme:

We want to find the **mathematical conditions** when all firms are profit maximizing

If all firms are profit maximizing, no firm wants to change its strategy – so we have a Nash Equilibrium!

Our mathematical conditions

	Firm 1	Firm 2
Cournot game	Profit is a function of q_1 and q_2 Maximise with respect to q_1 Find best response function: $q_1 = A - Bq_2$	Profit is a function of q_1 and q_2 Maximise with respect to q_2 Find best response function: $q_2 = A - Bq_1$
Bertrand game	Profit is a function of p_1 and p_2 Maximise with respect to p_1 Find best response function: $p_1 = A + Bp_2$	Profit is a function of p_1 and p_2 Maximise with respect to p_2 Find best response function: $p_2 = A + Bp_1$
Stackelberg game with Firm 1 going first	Profit is a function of q_1 only! (Substituting in Firm 2's BR function) Maximise with respect to q_1	Profit is a function of q_1 and q_2 Maximise with respect to q_2 Find best response function: $q_2 = A - Bq_1$



	Casino: Red	Casino: Black
	Probability that roulette game chooses red = p_C	Probability that roulette game chooses red = $(1 - p_C)$
Gambler: Red Probability that gambler chooses red = p_G	C = -1 G = 1	C = 1 G = -1
Gambler: Black Probability that gambler chooses black = $(1 - p_G)$	C = 1 G = -1	C = -1 G = 1